That is, $f'(x) = 0$.

or $\boxed{\frac{d}{dx}(k)=0}$.

**(2) The power function $y = x^n$, $n > 0$ is an integer.**

Let $f(x) = x^n$.

Then, $f(x+\Delta x) = (x+\Delta x)^n$ and

$f(x+\Delta x) - f(x) = (x+\Delta x)^n - x^n$.

This implies, $\dfrac{f(x+\Delta x)-f(x)}{\Delta x} = \dfrac{(x+\Delta x)^n - x^n}{(x+\Delta x)-x}$

This implies, $\lim\limits_{\Delta x\to 0}\dfrac{f(x+\Delta x)-f(x)}{\Delta x} = \lim\limits_{\Delta x\to 0}\dfrac{(x+\Delta x)^n - x^n}{(x+\Delta x)-x}$

$= \lim\limits_{y\to x}\dfrac{y^n - x^n}{y-x} = nx^{n-1}$, where $y = x+\Delta x$ and $y \to x$ as $\Delta x \to 0$.

That is, $\dfrac{d}{dx}f(x) = nx^{n-1}$ or $\boxed{\dfrac{d}{dx}(x^n)=nx^{n-1}}$.

**Corollary 10.1**

When $n=\dfrac{p}{q},(p,q)=1$, $\dfrac{d}{dx}\left(x^{\frac{p}{q}}\right)=\dfrac{p}{q}x^{\frac{p}{q}-1}$.

**Corollary 10.2**

For any real number $\alpha$, $\dfrac{d}{dx}(x^\alpha)=\alpha x^{\alpha-1}$.

For instance,

(1) $\dfrac{d}{dx}(5) = 0$ since 5 is a constant.

(2) $\dfrac{d}{dx}(x^3) = 3x^2$, by power function rule.

(3) $\dfrac{d}{dx}\left(x^{\frac{3}{2}}\right) = \dfrac{3}{2}x^{\frac{3}{2}-1} = \dfrac{3}{2}x^{\frac{1}{2}}$, by power function rule.

(4) $\dfrac{d}{dx}\left(x^{\sqrt{2}}\right) = \sqrt{2}x^{\sqrt{2}-1}$, by power function rule.

(5) $\dfrac{d}{dx}\left(x^{\frac{2}{3}}\right) = \dfrac{2}{3}x^{\frac{2}{3}-1} = \dfrac{2}{3}x^{\frac{-1}{3}}, (x\neq 0)$, by power function rule.

(6) $\dfrac{d}{dx}(100x^9) = 100\dfrac{d}{dx}(x^9) = 100\times 9x^{9-1} = 900x^8$ by theorem 10.6.



10_11th-Maths-Vol2-Unit10.indd 152 16-12-2022 11:05:09